

AI Automation with n8n

Lesson 6: Automating Content Creation

Automating Content Creation – Study Notes

Key Concepts

1. **Prompt driven outline generation** – Using AI prompts to produce structured blog outlines quickly.
2. **Article summarization** – Condensing long texts into concise summaries while preserving main ideas.
3. **Google Docs API integration** – Programmatically creating, editing, and saving documents in Google Drive.
4. **Workflow automation** – Linking the three steps (outline → summary → save) with tools like Zapier, Make (Integromat), or simple scripts.
5. **Prompt engineering basics** – Crafting clear, specific prompts to get reliable, repeatable outputs.

Summary

In this beginner lesson you learned how to automate three common content creation tasks: generating a blog outline, summarizing an article, and storing the result in Google Docs. The process starts with a well-crafted prompt that tells the language model what kind of outline you need (e.g., headings, sub-headings, word count hints). Next, you feed a source article (or its URL) into a summarization prompt that extracts the core points in a short paragraph or bullet list. Finally, you use the Google Docs API (or a no-code connector like Zapier) to create a new document, insert the outline and summary, and save it to a chosen folder in Google Drive. By chaining these steps together—either with a simple Python script or a visual automation platform—you can produce ready-to-publish content with minimal manual effort.

The lesson also covered best practices for prompt engineering (be explicit, give format instructions, set length limits) and basic authentication with Google's OAuth 2.0 flow, so you can securely connect your script to your Google Workspace account. With these building blocks, you can expand the workflow to include additional steps such as adding images, SEO metadata, or publishing directly to a CMS.

Important Points

- **Prompt clarity:** State the desired structure, tone, and length. Example: “Create a 5-point outline for a 1,200-word blog post about remote work productivity, using H2 headings.”
- **Summarization style:** Choose **extractive** (pull key sentences) or **abstractive** (re-write in new words) depending on the use case.
- **Google Docs API basics:**
 - Enable the Docs and Drive APIs in Google Cloud Console.
 - Use OAuth 2.0 credentials (client ID/secret) for user auth or a service account for automated scripts.
 - Key endpoints: `documents.create`, `documents.batchUpdate`, `files.create` (Drive).
- **Rate limits & quotas:** Google APIs have daily quotas; batch updates reduce the number of calls.

- **Automation tools:**

- **Zapier/Make:** Drag and drop triggers (e.g., new outline in a Google Sheet) !' actions (summarize via OpenAI !' create Doc).

- **Python script example:** ``openai.ChatCompletion.create()` !' `googleapiclient.discovery.build('docs', 'v1')`.`

- **Error handling:** Check for API errors (401, 429) and implement retries with exponential back off.

Examples

Example 1 – Python script that creates a blog outline, summarizes an article, and saves to Google Docs

```
```python
```

```
import os, openai, json
```

```
from google.oauth2 import service_account
```

```
from googleapiclient.discovery import build
```

#### ----- 1. Generate outline -----

```
outline_prompt = """
```

```
Create a detailed outline for a 1500 word blog post titled
```

```
"How AI is Transforming Small Business Marketing".
```

```
Include H2 headings and 2 3 bullet points under each heading.
```

```
"""
```

```
outline = openai.ChatCompletion.create(
```

```
 model="gpt-4o-mini",
```

```
 messages=[{"role":"user","content":outline_prompt}],
```

```
 temperature=0.3,
```

```
).choices[0].message.content
```

#### ----- 2. Summarize an article -----

```
article_text = open("article.txt").read()
```

```
summary_prompt = f"""
```

```
Summarize the following article in 5 concise bullet points.
```

```
Article:
```

```
{article_text}
```

```
"""
```

```
summary = openai.ChatCompletion.create(
```

```
 model="gpt-4o-mini",
```

```
 messages=[{"role":"user","content":summary_prompt}],
```

```
 temperature=0.2,
```

```
).choices[0].message.content
```

#### ----- 3. Save to Google Docs -----

```

SCOPES = ['https://www.googleapis.com/auth/documents',
 'https://www.googleapis.com/auth/drive.file']
creds = service_account.Credentials.from_service_account_file(
 "service_account.json", scopes=SCOPES)
docs_service = build('docs', 'v1', credentials=creds)
drive_service = build('drive', 'v3', credentials=creds)

```

## Create a blank document

```

doc = docs_service.documents().create(body={"title": "AI Marketing Blog Draft"}).execute()
doc_id = doc.get('documentId')

```

## Prepare content

```

content = f"## Outline\n{outline}\n\n## Summary\n{summary}"
requests = [{
 "insertText": {
 "location": {"index": 1},
 "text": content
 }
}]
docs_service.documents().batchUpdate(documentId=doc_id, body={"requests": requests}).execute()
print(f"Document created: https://docs.google.com/document/d/{doc_id}")

```

**\*Explanation\*:** The script first asks GPT 4o mini for an outline, then for a 5 point summary of a local article file, and finally uses a service account credential to create a Google Doc and insert the generated text.

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### Example 2 – No code Zapier workflow

Zap Trigger	Zap Action 1	Zap Action 2	Zap Action 3
**\*\*New row in Google Sheet\*\*** (columns: **\*Title**, **\*URL\***)	**\*\*OpenAI – “Create Outline”\*\*** (prompt uses the **\*Title\*** field)	**\*\*OpenAI – “Summarize URL”\*\*** (fetches article from **\*URL\***)	**\*\*Google Docs – “Create Document from Text”\*\*** (combines outline + summary, saves to folder **\*Content Drafts\***)

**\*Explanation\*:** When you add a new row with a blog idea, Zapier automatically generates an outline, fetches and summarizes the linked article, and drops everything into a ready to edit Google Doc.

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## Quick Review

- \*\*What two main types of summarization exist, and when might you choose each?\*\***
- \*\*Name the three essential steps in the automated content creation workflow covered in this lesson.\*\***
- \*\*Which Google APIs are required to create and edit a document programmatically?\*\***
- \*\*Give one tip for writing an effective prompt that produces a clean blog outline.\*\***
- \*\*If you hit a 429 error while calling the Docs API, what should you do?\*\***

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## Further Reading

- **Prompt Engineering** – “Effective Prompt Design for Large Language Models” (OpenAI Cookbook).
- **Google Docs API Deep Dive** – Official documentation, especially the `batchUpdate`` request format.
- **Automation Platforms** – Compare Zapier, Make, and n8n for content pipelines.
- **SEO friendly Content Generation** – Using AI to suggest keywords, meta descriptions, and internal linking.
- **Ethical Considerations** – Attribution, plagiarism detection, and responsible AI use in publishing.

